

GEOG 28IA

Data Models, Databases, and Their Limits

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Learning Objectives: To understand the primary data models used in GIS, and how they relate to common database structures. To apply those structure to 'non-traditional' conceptualizations of landforms



GOOD OMENS II

THE G IN GISCIENCE

PROMPT

Has GIScience drifted from Geography's disciplinary ambitions (think 200B)?
Has big data, AI, data-driven investigation quickened or slowed that drift?

PROMPT

Are you worried about the G in GIScience? What have you seen in the conferences and research talks you have attended?

BREAK

DEFINING PLACE

ACTIVITY

Purpose: Ensure a common understanding of the major data models of GIS

Task: Together, we will outline how the vector and raster data models represent space in a GIS.

STUDENT PERSPECTIVE

The readings implicitly separate three layers of spatial representation as geometric representation, data management, and conceptual categorization but what happens when these layers conflict? For example, what happens when human perceived space does not align with geometric or topical representation.

BREAK

MARK AND TURK (2003)

STUDENT PERSPECTIVE

Can process-oriented work be integrated by incrementally extending existing GIS, or does meaningful progress require rethinking what stays at the disciplinary core?

ACTIVITY

Purpose: Apply your understanding of GIS data models and spatial ontologies to the challenge of representing different conceptualizations of landforms in a GIS

Task: Work together to propose an approach to representing the Yindjibarndi conceptions of water features and landforms, as presented in Mark and Turks (2023), in a GIS.

- Be as radical as you like in your approach
- Suggestion: Break down the concepts presented. What needs to be represented? What are your options?
- If you say this cannot be done, or should not be done, develop an argument as to why.

STUDENT PERSPECTIVE

Instead of dividing time in a spatial way of past, present and future, this brings me a different conceptualization. Putting time in an absolute, relative and interrelated way. Endurance talks about how long substance has its form, this is in an absolute and relative way. Perdurant talks about interrelationship between substances. In this way, it emphasizes process and time, so the substance might not have concrete form.

As the Chinese philosopher might say: “There are two types of things, one has constant form while the other has constant principle, like water and cloud.” If we are applying this conceptualization into GISystems, the emphasis of the design would be dependent on process and time, instead of forms, like Mark’s paper suggests.

OPEN QUESTION

How do we avoid moving from the rigidity of formalism to the opposite problem of arbitrariness and impressionistic interpretation? In other words, when these materials criticize existing GIS representations, are they really pointing toward a more rigorous scientific program, or mainly toward a broader philosophical unease with formal representation itself?